Milestone 3 Report

Smart MCQ Solver Challenge

Deep Learning and Generative AI Project

Author: Vishal Sharma

---

1. Introduction

Milestone 3 introduces Retrieval-Augmented Generation (RAG), a framework that combines information retrieval with transformer-based reasoning. Instead of relying solely on a language model's internal knowledge, relevant documents are first retrieved from an external knowledge base and incorporated into the model's input. This approach enhances reasoning accuracy by providing additional contextual information.

---

2. Objectives

The primary objectives of this milestone are:

- Understand Retrieval-Augmented Generation.
- Generate dense document embeddings.
- Build a vector database using FAISS.
- Retrieve semantically similar documents.
- Improve retrieval quality using Cross-Encoder reranking.
- Construct context-augmented prompts.
- Evaluate retrieval effectiveness.
- Analyze the impact of incorrect retrieved context.

---

3. Knowledge Base Construction

A knowledge base was created by converting documents into dense vector representations using the `all-MiniLM-L6-v2` Sentence Transformer model.

Each document was embedded into a fixed-length numerical representation suitable for semantic search.

---

4. FAISS Vector Database

Facebook AI Similarity Search (FAISS) was used to efficiently index dense embeddings.

The FAISS index enables fast nearest-neighbor retrieval even for large document collections.

Key operations include:

- Index creation
- Embedding insertion
- Similarity search
- Top-k retrieval

---

5. Semantic Retrieval

For each query prompt:

1. Generate embedding.
2. Search the FAISS index.
3. Retrieve the top-k most similar documents.

Retrieved documents provide external knowledge for downstream reasoning.

---

6. Cross-Encoder Re-ranking

Although FAISS retrieves relevant documents efficiently, retrieved rankings may not always be optimal.

A Cross-Encoder (`cross-encoder/ms-marco-MiniLM-L-6-v2`) was applied to score the retrieved documents jointly with the query.

The reranked documents provide higher semantic precision than retrieval alone.

---

7. Retrieval-Augmented Generation

Retrieved documents were concatenated with the original question to create augmented prompts.

The prompt format used was:

```
Context: [Retrieved Documents]

Question: [Prompt]
```

The augmented prompt was then processed using a transformer-based classifier.

---

8. Zero-Shot Classification

The augmented prompts were evaluated using the `facebook/bart-large-mnli` zero-shot classification model.

Predicted probabilities before and after retrieval augmentation were compared to evaluate the benefits of incorporating external context.

---

9. Adversarial RAG

To study retrieval robustness, intentionally incorrect documents were supplied as context.

This experiment demonstrated that poor retrieval quality can negatively impact downstream model predictions, highlighting the importance of accurate retrieval systems.

---

10. Retrieval Evaluation

Retrieval quality was evaluated using Hit Rate.

A retrieval was considered successful if the correct document appeared among the retrieved top-k results.

This metric provides an intuitive measure of retrieval effectiveness.

---

11. Results

The milestone successfully implemented:

- Dense embedding generation
- FAISS vector indexing
- Top-k semantic retrieval
- Cross-Encoder reranking

- Context augmentation
- Zero-shot reasoning with retrieved knowledge
- Adversarial retrieval analysis
- Retrieval Hit Rate evaluation

These experiments demonstrate the practical advantages of combining retrieval systems with transformer models.

---

12. Challenges

Several practical challenges were encountered during implementation:

- Selecting appropriate embedding models.
- Managing FAISS index construction.
- Balancing retrieval speed and accuracy.
- Computational overhead of Cross-Encoder reranking.
- Maintaining relevant contextual information.
- Evaluating retrieval robustness under adversarial settings.

These challenges illustrate the trade-offs involved in designing effective RAG systems.

---

13. Future Work

The next milestone focuses on transformer fine-tuning using LoRA and supervised learning techniques.

Upcoming tasks include:

- MCQ data formatting
- LoRA fine-tuning
- Full transformer fine-tuning
- Training optimization
- GPU memory management
- Efficient batch processing

---

14. Conclusion

Milestone 3 successfully implemented a complete Retrieval-Augmented Generation (RAG) pipeline. By integrating dense embeddings, FAISS retrieval, Cross-Encoder reranking, and context-aware reasoning, the system demonstrated improved semantic retrieval capabilities compared to standalone transformer inference. These techniques establish the retrieval foundation required for fine-tuned question-answering models in subsequent milestones.